

Geophysical Equations and Field Operators

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1 What is Geophysics?

Geophysics is the study of the physical properties of the Earth (and other planetary bodies) and the physical processes that operate on or within them, using observations, modeling, and controlled experiments on physical fields (??). A *field* is any quantity that can be specified throughout space (and possibly time), with a position and a value (magnitude and, for vectors, direction).

Examples of fields

- Scalar fields: temperature $T(\mathbf{x})$, elevation $z(\mathbf{x})$, composition variables (e.g., modal fraction) (??).
- Vector fields: gravitational acceleration $\mathbf{g}(\mathbf{x})$, wind velocity $\mathbf{u}(\mathbf{x})$, magnetic field $\mathbf{B}(\mathbf{x})$.
- Tensor fields: Cauchy stress $\boldsymbol{\sigma}(\mathbf{x})$, strain $\boldsymbol{\epsilon}(\mathbf{x})$.

A field can be represented as a function $f(\mathbf{x})$ mapping each position $\mathbf{x} = (x, y, z)$ to a value in $\mathbb{R}$, $\mathbb{R}^3$, or a tensor space depending on type.

2 Describing Change: Derivatives and the Gradient

For a scalar field $f(\mathbf{x})$, the rate of change along coordinate directions is given by partial derivatives:

$$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}. \quad (1)$$

The *gradient* of f is the vector field

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right). \quad (2)$$

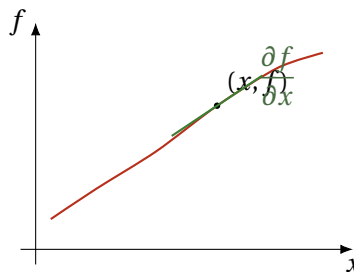


Figure 1: Slope of a scalar field along x at a point (schematic).

3 Divergence: Measuring Outflow of a Vector Field

For a vector field $\mathbf{F} = (F_x, F_y, F_z)$, the *divergence* is

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}. \quad (3)$$

Physically, $\nabla \cdot \mathbf{F}$ measures the net “density change” or source strength at a point. A classic example is a radially expanding flow (e.g., an explosion), which has positive divergence away from the source.

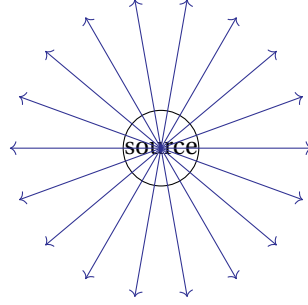


Figure 2: Schematic of a diverging vector field (explosive-like outflow).

4 Divergence Theorem and the Laplacian

Over a volume V with boundary surface ∂V and outward unit normal $\hat{\mathbf{n}}$, the divergence theorem states

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \iint_{\partial V} \mathbf{F} \cdot \hat{\mathbf{n}} dS. \quad (4)$$

The *Laplacian* of a scalar field is the divergence of its gradient:

$$\nabla^2 f \equiv \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}. \quad (5)$$

It measures curvature of the field and appears ubiquitously in geophysical governing equations.

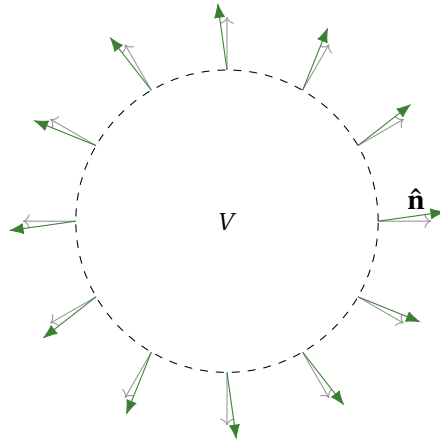


Figure 3: Flux through a closed surface ∂V equals volume integral of $\nabla \cdot \mathbf{F}$ (schematic).

5 Canonical Field Equations

5.1 Poisson and Laplace Equations

Many static (time-independent) geophysical fields satisfy Poisson's equation

$$\nabla^2 \Phi(\mathbf{x}) = S(\mathbf{x}), \quad (6)$$

where Φ is a potential and S is a source term determined by the underlying physics. In source-free regions ($S = 0$), this reduces to Laplace's equation

$$\nabla^2 \Phi = 0. \quad (7)$$

5.2 Examples

- Gravitation (Newtonian, scalar potential Φ_g): $\nabla^2 \Phi_g = 4\pi G \rho$. (units “fudge factor” comment in notes: ??)
- Steady heat conduction (temperature T with volumetric heat production A and conductivity k): $\nabla^2 T = -\frac{A}{k}$ (sign/convention in notes: ??).
- Magnetostatics (scalar magnetic potential under curl-free conditions): form depends on magnetization $\mathbf{M}$ and permeability μ (exact form in notes: ??).

6 Diffusion

Time-dependent smoothing (diffusive) processes satisfy

$$\frac{\partial u}{\partial t} = \kappa \nabla^2 u + S(\mathbf{x}, t), \quad (8)$$

with diffusivity κ (“diffusion constant” in notes). In geothermics, $u \equiv T$ and κ is the thermal diffusivity (the notes suggest we may not use this often here: ??).

7 Constitutive Relations (Statics)

- Electrical conduction (Ohm's law): $\mathbf{J} = \sigma \mathbf{E}$.
- Magnetism: $\mathbf{B} = \mu \mathbf{H}$; with magnetization, $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$ (specific form referenced in notes: ??).

8 Wave Phenomena

Propagating fields typically satisfy a second-order wave equation

$$\frac{\partial^2 u}{\partial t^2} = v^2 \nabla^2 u, \quad (9)$$

where v is the propagation speed.

Seismology. In elastic media, compressional (P) and shear (S) waves have speeds v_p and v_s respectively (details and exact parameter relations deferred in the notes: ??).

9 Electromagnetics (constants and vacuum relations)

The speed of light in vacuum is related to the permittivity and permeability of free space by

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}. \quad (10)$$

Further wave equations for $\mathbf{E}$ and $\mathbf{H}$ in free space (if needed in these notes: ??).

10 Fundamental Quantities

- **Mass** m : amount of matter (units: kg).
- **Position** $\mathbf{x} = (x, y, z)$ with unit vectors $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ (units: m). Magnitude $r = \|\mathbf{x}\|$.
- **Velocity** $\mathbf{v} = d\mathbf{x}/dt$ (units: m s^{-1}).
- **Acceleration** $\mathbf{a} = d\mathbf{v}/dt = d^2\mathbf{x}/dt^2$ (units: m s^{-2}).
- **Force** $\mathbf{F}$: interaction that changes motion. Newton's 2nd law: $\mathbf{F} = m\mathbf{a}$ (units: $\text{N} = \text{kg m s}^{-2}$).

Static equilibrium. At rest does not imply “no forces”; rather, forces balance (e.g., gravity vs. normal reaction at the ground), so net force is zero: $\sum \mathbf{F} = \mathbf{0}$.

11 Work and Energy

Work by a force along a path $\mathcal{C}$: $W = \int_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{s}$ (units: $\text{J} = \text{N m}$). The dot product projects the force along the direction of motion. Kinetic energy $K = \frac{1}{2} m v^2$.

Work–energy relation. Using $dK/dt = \mathbf{F} \cdot \mathbf{v}$, integration along the path gives the change in kinetic energy equals work done.

12 Potential and Path Independence

When the work between two points depends only on endpoints (not the path), the field is conservative and admits a potential Φ with $\mathbf{F} = -\nabla\Phi$ (sign conventions in the notes mention “force is gradient of work”; adopt $\mathbf{F} = -\nabla\Phi$ here and reconcile with class convention: ??).

$$W_{P \rightarrow Q} = \int_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{s} = -[\Phi(Q) - \Phi(P)]. \quad (11)$$

Differential form: $\nabla \times \mathbf{F} = \mathbf{0}$. Integral form: line integral is path independent.

13 Field Types and Representations

- **Scalar field** $T(\mathbf{x})$ (e.g., temperature, elevation).
- **Vector field** $\mathbf{g}(\mathbf{x}), \mathbf{B}(\mathbf{x})$.
- **Tensor field** $\sigma(\mathbf{x}), \epsilon(\mathbf{x})$.

13.1 Visualizations

Quiver plot (vectors).

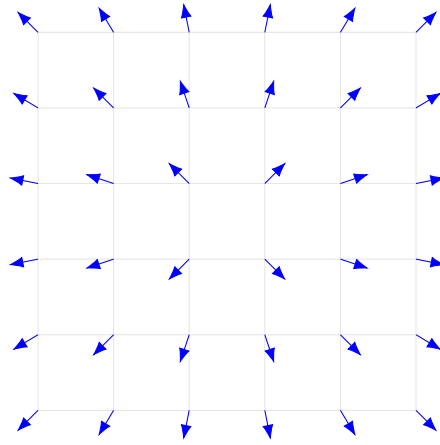


Figure 4: Quiver depiction of a simple radial vector field.

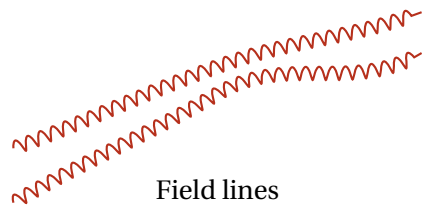


Figure 5: Sketch of field/force lines.

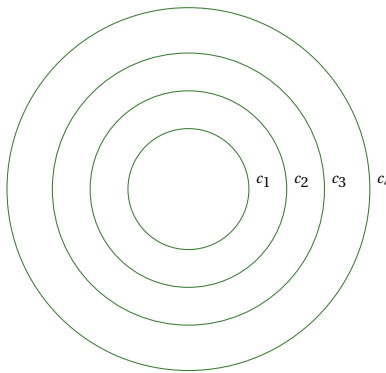


Figure 6: Iso-contours (perpendicular to field lines in conservative fields).

Field lines.

14 Continuity and Differentiability

A function is continuous if it can be traced without breaks. Differentiability additionally requires the slope to be well-behaved (left and right derivatives agree). Points with corners or cusps are non-differentiable.

15 Superposition

In linear systems, fields add. For sources S_1, S_2 , the resulting field is $f = f_1 + f_2$ (additivity) and scales with source strength (homogeneity). This principle underlies many constructions in potential theory.

$$f(\alpha S) = \alpha f(S), \quad (12)$$

$$f(S_1 + S_2) = f(S_1) + f(S_2). \quad (13)$$

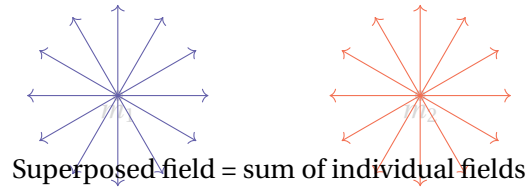


Figure 7: Illustration of superposition from two sources.

Transcription notes: Where the original handwriting or OCR was ambiguous, a “??” marker should be inserted during detailed equation derivations (none were strictly required in this summary; add if needed upon your review). Diagrammatic sketches have been redrawn schematically in TikZ.